

Hilda Adwubi Osei

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Research Interests

- Themes: Robustness under distributional shift, multi-agent systems and information asymmetry, fairness in algorithmic and resource allocation.
- Methods: Convex optimization, agent-based and game-theoretic simulation, statistical and predictive modeling (regression, ensemble methods, transfer learning).
- Applications: Agricultural markets and economics, public health (infection prevention, environmental health).

Education

Kwame Nkrumah University of Science and Technology
B.Sc. Industrial Engineering

Kumasi, Ghana
2021 – 2024

Publications

Work in Progress

WP1. Osei, H.A. *FarmDirect: Data-Driven pricing and risk-adjusted allocation for Ghanaian tomato*. Manuscript submitted to arXiv (pending moderation review, cross-disciplinary classification); targeting journal submission.

Conference and Poster Presentations

CP1. Osei, C.T., Asare, A.O., Agyen, Y.O., Osei, H.A., Amooba, P.A. *Knowledge and Compliance with Standard Precautions for Nosocomial Infection Prevention among Undergraduate Nursing and Midwifery Students at a Ghanaian University*. Poster presentation, Korle Bu Teaching Hospital, Ghana, 2026.

Preprints and Technical Reports

P2. Osei, H.A. *Learning from waste: Machine Learning for health risk prediction and computer vision-based sorting in Ghana*. arXiv:2608.25759 [cs.LG] (cross-listed cs.CV, cs.CY), 2026.

P1. Osei, C.T., Asare, A.O., Agyen, Y.O., Osei, H.A., Amooba, P.A. *Knowledge and Compliance with Standard Precautions for Nosocomial Infection Prevention among Undergraduate Nursing and Midwifery Students at a Ghanaian University*. medRxiv, 2026.

Research and Work Experience

FarmDirect

Founder & Lead Developer

Accra, Ghana
2025 – Present

- Designing a data-driven marketplace to connect farmers directly to buyers, eliminating middlemen and optimizing prices; conducted the underlying research through predictive modeling, risk-adjusted optimization, and agent-based simulation on real government market data.
- Selected as 1 of 9 finalists to pitch before a national judging panel, winning an Innovation Grant (GHS 10,000); presented at the “We, The Youth” Global Youth Summit (South Korea, 1 of 150 selected from 27,000+ applicants).

Ghana Atomic Energy Commission (GAEC)

Research & Innovation Collaborator

Accra, Ghana
Jun 2025 – Present

- Collaborate on a funded project applying technology to support farmers, building on research developed through FarmDirect.

Headstarter AI
Software Engineering Fellow

Remote
Jul – Sep 2024

- Built AI-powered web applications in agile sprint cycles as part of a remote fellowship cohort.

University Service of Canada / CTVET, National Software Development Summer School Ghana
Fellow, then Mentor *Apr – Aug 2025 (Fellow); 2025 – Present (Mentor)*

- Ranked among the top graduates of the program; selected as 1 of 9 finalists to pitch before national judges.
- Now mentors the following cohort of the same national summer school.

Teaching Experience

Independent Technology Education Initiative
Programming Mentor

Ghana
2026

- Invited to teach Python programming to high school students.

National Software Development Summer School, CTVET/GAEC
Teaching Assistant

Ghana
2025

- Mentor incoming fellows of the national summer school program following completion as a top-ranked graduate and national pitch finalist in the same program.

MATH 152: Calculus with Analysis, Kwame Nkrumah University of Science and Technology Kumasi, Ghana
Teaching Assistant *2024*

- Graded assignments and exams, led tutorials, and held office hours for students in the course.

ME 259: Applications of Computer Graphics, Kwame Nkrumah University of Science and Technology Kumasi, Ghana
Teaching Assistant *2022*

- Graded assignments and exams, led tutorials, and held office hours for students in the course.

Honors and Awards

- IFWY Global Explorer, International “We, The Youth” Forum, South Korea, 2025.
- Innovation & Sustainability Finalist, Mastercard Foundation x GAEC Challenge, 2025.
- Best Innovation Award, National Software Development Bootcamp, Ghana, 2025.
- GAEC & Mastercard Foundation Research Collaboration Recognition, 2025.
- Software Engineering Fellowship, Headstarter AI, 2024.
- STEM Women Ghana Changemaker Award, 2024.
- Academic Distinction, Second Class Upper, KNUST, 2024.
- TechWomen Africa Rising Recognition, 2023.
- Youth Arise Leadership Fellowship & Counselor Appointment, 2020.

Invited Talks

- Economic Transformation and Youth Inclusion. International “We, The Youth” Forum, UNRISD, Botswana, 2025.
- Economic Transformation and Youth Inclusion. International “We, The Youth” Forum, UNRISD, South Korea, 2025.
- Youth Engagement in Climate-Induced Loss and Damage in Agrifood Systems. World Food Forum, 2025.